

Threshold re-check: a test bug changed the answer

A corrected re-run of the independence test on the 28 enrolled La Salle students

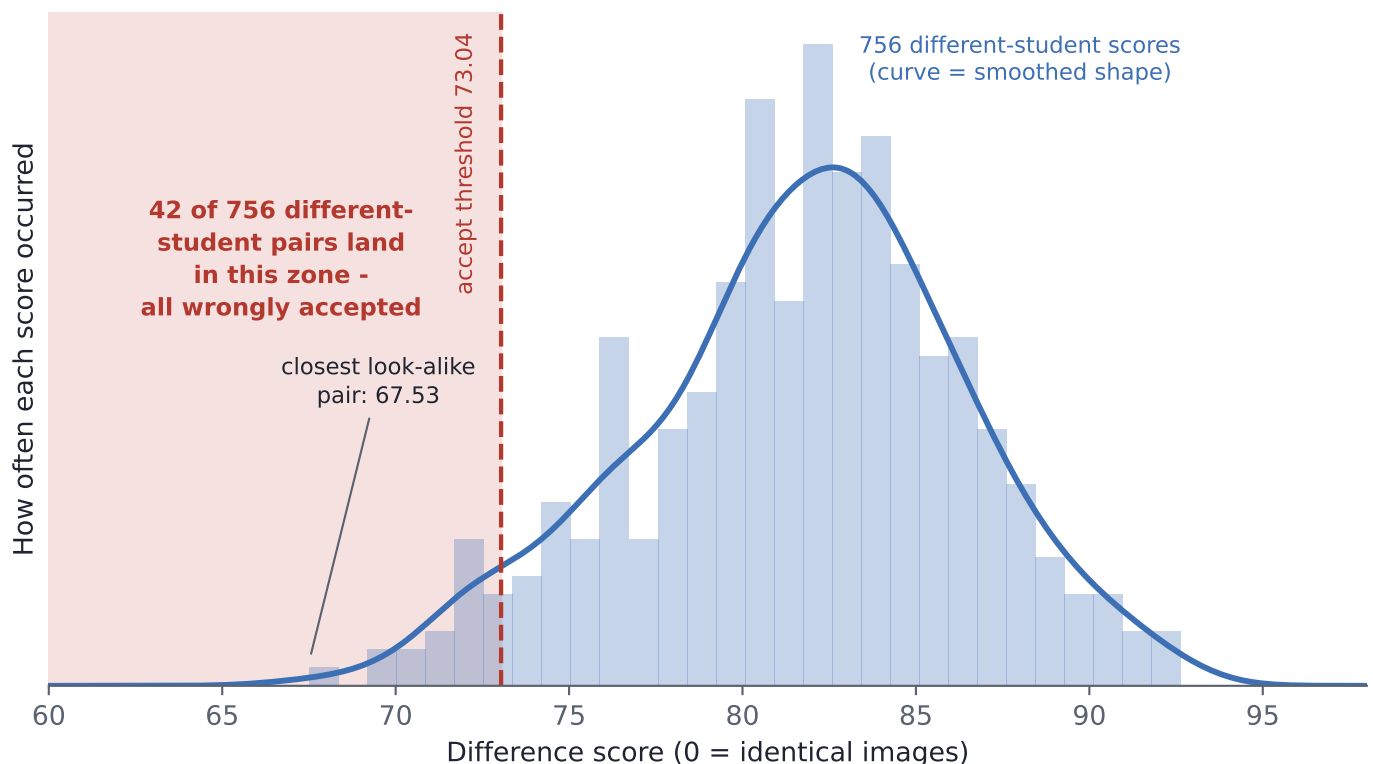
Corrected answer: our first report was too optimistic. A fixed setting matters.

The earlier version of this report picked one RANDOM photo per student, which happened to make our texture check look perfect (0 mistakes). Re-run properly - the same standard photo for every student, confirmed 3 times independently - it wrongly matches 42 of 756 different-student pairs.

The test in one paragraph

We took 28 enrolled students, one standard photo each (same lighting, same pose - the photo everyone's ID would use), and compared every photo against every other student's: 756 comparisons, every one of them two different people. Our texture check (the fast first-pass scan) says "same person" whenever the two photos score 73.04 or less on its difference scale. Since every pair here really is two different people, any acceptance is a mistake. Result: 42 mistakes out of 756 (about 1 in 18) - not zero. Our second, independent check (a deep-learning face matcher) catches most of those mistakes when the first check is unsure, bringing the full system's mistakes down to 12 of 756 (about 1 in 63).

The result: a real chunk of different-student pairs falls inside the accept zone



Why the first report missed this

Our student photos come in 12 conditions (bright/dim lighting x six angles). The first report let the computer pick a random condition per student, so any two students being compared were usually also compared across different lighting and angle - an extra, artificial source of "difference" that had nothing to do with the students' actual faces, and that inflated the scores enough to hide real look-alikes. Locking everyone to the same standard photo condition removes that artificial boost and shows the texture check's true, weaker performance on faces alone.

Two checks are better than one - but don't fully close the gap

Our system never relies on the texture check alone: any uncertain case gets a second opinion from a different-technology deep-learning check before it's accepted. Here is what each layer does alone, and combined, on the corrected 756-comparison test:

Layer	Wrong matches	Rate	What it means
Texture check alone (threshold 73.04)	42 of 756	5.56%	The weak link exposed by the fixed-photo re-test.
Deep-learning check alone	14 of 756	1.85%	Different technology, different mistakes - only 4 overlap.
Both checks wrongly agree	4 of 756	0.53%	The failure the combined system cannot catch.
Full system (combined, as deployed)	12 of 756	1.59%	The second check catches most, not all, of the texture leak.

The 100-per-million target came from testing against a large public photo set (LFW), not our own students. Our own students - who look more alike as a group (classmates, similar age, same standard photo booth) - are a harder crowd for the texture check than that public benchmark predicted.

What changed since the last version of this report

What we measured	First report (retracted)	Corrected report
Texture check wrong matches	0 of 756 (bugged)	42 of 756 (5.56%)
Deep-learning check wrong matches	20 of 756 (2.65%)	14 of 756 (1.85%)
Full system wrong matches	9 of 756 (1.19%)	12 of 756 (1.59%)
Closest look-alike score seen	73.22	67.53

The corrected numbers are confirmed 3 times independently (bit-for-bit identical each time) using the same standard photo per student. The deployed threshold value (73.04) itself did not change - only the test methodology used to check it did.

Bottom line

The professor's concern was right, for a reason our first report missed: a test-selection bug (a random photo per student instead of one standard photo for everyone) made the texture check look flawless when it was not. Retested properly, the texture check alone lets through 1 in 18 look-alike pairs among our own students - well above what the public-benchmark calibration promised. Our two-check design already cuts that down to about 1 in 63, which is real protection, but not yet at our original target. Next step: re-calibrate the threshold using our own students' data instead of only the public benchmark, and always test with one fixed photo condition going forward.